| Program: **Diploma in Integrated Circuit Design and Fabrication** | |
|---|---|
| Course Code : **4373** | Course Title: **CAD for VLSI** |
| Semester: **4** | Credits: **3** |
| Course Category: **Program Core** | |
| Periods per week: **3 (L:3 T:0 P:0)** | Periods per semester: **45** |

**Course Objectives:**

After studying this course, students will be able to:

- Understand the fundamentals of VLSI CAD design.
- Understand the use of graph algorithms.
- Understand floor planning, placement and routing.
- Understand the use of advanced techniques in VLSI layout design.

**Course Prerequisites:**

| Topic | Course code | Course name | Semester |
|---|---|---|---|
| VLSI design flow | | VLSI design | 4 |

**Course Outcomes:**

On completion of the course, the student will be able to:

| COn | Description | Duration (Hours) | Cognitive level |
|---|---|---|---|
| CO1 | Explain various CAD tools and Overview of commonly used data structures | 10 | Understanding |
| CO2 | Explainthe concepts of various algorithms and methods of floor planning used for VLSI design | 11 | Understanding |
| CO3 | Explainthe concepts of placement and routing algorithms used for VLSI Design | 11 | Understanding |
| CO4 | Discuss the Advanced Topics in VLSI CAD | 11 | Understanding |
| | Series Test | 2 | |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 3 | | | | | | |
| **CO2** | 3 | | | | | | |
| **CO3** | 3 | | | | | | |
| **CO4** | 3 | | | | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline**

<table>
<tr><th>Module Outcomes</th><th>Description</th><th>Duration (Hours)</th><th>Cognitive Level</th></tr>
<tr><td>CO1</td><td colspan="3">Explainvarious CAD tools and Overview of commonly used data structures</td></tr>
<tr><td>.01</td><td>Discuss the role of VLSI CAD tools in design process</td><td>2</td><td>Understanding</td></tr>
<tr><td>M1.02</td><td>Illustrate the terms layout, routing and placement</td><td>2</td><td>Understanding</td></tr>
<tr><td>M1.03</td><td>Explain EDA tools and Cad software used in VLSI design</td><td>3</td><td>Understanding</td></tr>
<tr><td>M1.04</td><td>Explain the common used data structures used in VLSI CAD</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents: Introduction to CAD for VLSI: Basic Terminology in VLSI CAD-Overview of VLSI CAD tools, role in the VLSI design process, Definition of terms- layout, routing, placement, etc. Introduction to CAD Tools-Role in automating VLSI design flow, improving design productivity, explanation of Electronic Design Automation (EDA) tools, CAD software used in VLSI design. Introduction to Data Structures in CAD-Definition of data structures, role in CAD, Overview of commonly used data structures in VLSI CAD, Role in efficient design representation, Linked List and its Applications in Layout Editors.</td></tr>
<tr><td>CO2</td><td colspan="3">Explainthe concepts of various algorithms and methods of floor planning used for VLSI design</td></tr>
<tr><td>M2.01</td><td>Discuss the applications of DFS and BFs in graph connectivity</td><td>2</td><td>Understanding</td></tr>
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| | | | |
|---|---|---|---|
| M2.02 | Explain the Properties and applications of spanning tree in circuit layout and design optimization | 3 | Understanding |
| M2.03 | Illustrate shortest path algorithms | 3 | Understanding |
| M2.04 | Explain the floor planning methodologies in VLSI design | 3 | Understanding |
| | Series test -1 | 1 | |
| **Contents: Graph Algorithms and Floor Planning:** Introduction to Graphs and their Representations- Definition, components, and representations of graphs. Graph Algorithms for Searching and Traversal- DFS and BFS applications in graph traversal and connectivity. Introduction to Spanning Trees-Applications- Properties and applications in circuit layout and design optimization. Basics of Shortest Path Algorithms-Introduction to Dijkstra's and Bellman-Ford algorithms. Introduction to Floor Planning in VLSI Design-Objectives, methodologies, and constraint-based techniques. | | | |
| **CO3** | Explainthe concepts of placement and routing Algorithms used for VLSI Design | | |
| M3.01 | Explain the challenges and strategies for pin assignment | 2 | Understanding |
| M3.02 | Overview of placement algorithms-simulated annealing | 2 | Understanding |
| M3.03 | Discuss the importance of global routing problem and algorithms | 3 | Understanding |
| M3.04 | Explain Maze routing Algorithm and Lees algorithms | 4 | Understanding |
| **Contents: Placement and Routing:** Introduction to Pin Assignment in VLSI Design-Role, challenges, and strategies for pin assignment. Basics of Placement Algorithms- Overview of placement algorithms-simulated annealing. Introduction to Simulation-Based Placement Techniques-Explanation of annealing-based methods, evolutionary-based methods. Basics of Global Routing and its Importance-Understanding the global routing problem and algorithms. Introduction to Maze Routing Algorithms-Explanation, Lee's algorithm. | | | |
| **CO4** | Discuss the Advanced Topics in VLSI CAD | | |
| M4.01 | Explain partitioning Algorithm | 2 | Understanding |
| M4.02 | Discuss Neighbour Pointers and Corner-Stitching Techniques and their applications | 3 | Understanding |

| | | | |
|---|---|---|---|
| M4.03 | Explain Bin-based method for layout representation | 1 | Understanding |
| M4.04 | Illustrate Min-Cut and Max-Cut Algorithms | 2 | Understanding |
| M4.05 | Discuss the importance timing closure techniques in VLSI design | 3 | Understanding |
| | Series 2 | 1 | |

**Contents: Advanced Topics in VLSI CAD:** Classification of Pin Assignment Problems-Study of different types, complexity, and optimization approaches. Basics of Partitioning Algorithms-Explanation of algorithms and their use cases. Neighbour Pointers and Corner-Stitching Techniques-Understanding neighbour pointers and applications. Bin-Based Method for Layout Representation-Explanation and advantages. Introduction to Min-Cut and Max-Cut Algorithms-Definition and overview of algorithms for VLSI design. Timing Closure Techniques-Introduction to timing closure, its importance in VLSI design.

**Text / Reference:**

| T/R | Book Title/Author |
|---|---|
| T1 | VLSI CAD,Niranjan N.Manjunath Kotari Chiplunkar ,prentice Hall India Learning Pvt Ltd(2011) |
| T2 | ALGORITHMS FOR VLSI DESIGN AUTOMATION, **Sabih H. Gerez,** paperback edition (2006) from Wiley |
| R1 | Algorithm for VLSI physical design automation, Naveed sherwani,klouwer academic publishers |

**Online Resources:**

| Sl.No. | Website Link |
|---|---|
| 1 | https://chipedge.com/top-11-eda-tools-in-vlsi/ |
| 2 | https://www.baeldung.com/cs/minimum-cut-graphs |
| 3 | https://www.vlsiacademy.org |